Amalgamation, HNN extensions and Artin twists of coset geometries

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Abstract

We give complete applications of the common coset-intersection theory to the constructions in Piedade–Tranchida, arXiv:2505.24662v1. The free-product and coherent-amalgam theorems include their converses, residue behavior and exact panel indices. Indexed HNN extensions recover the firm and residually connected branches; an explicit nonfirm input disproves the unrestricted geometric conclusion. The semidirect and Artin constructions have exact presentation and parabolic adapters. A full-space reflection argument separates Shephard generators, and the affine example is treated through its actual quotient intersections, including its exceptional input parameter. Concrete examples and the limit of graph iteration are proved in the same conventions.

Throughout, "the source" means the exact 60-page arXiv v1 specified above. We use left cosets, finite type sets, and a specified action on indexed types. A maximal parabolic means the subgroup assigned to one type; it need not be a maximal proper subgroup. Faithfulness, firmness, residual connectedness and flag transitivity are separate properties. The common paper *Coset intersections under group operations* is a complete declared proof dependency, not an unpaid reference to a proof bank.

The target results and examples remain attributed to Claudio Alexandre Piedade and Philippe Tranchida. Literal false or incomplete statements are distinguished from the valid corrected conclusions. All 81 named source records have a separate exact disposition and migration map; that inventory does not claim to reprove every stronger unused auxiliary assertion. In particular the broad fundamental-geometry and categorical questions remain open within this edition.

1 Free products and coherent amalgamation

The common free-product theorem gives N005 Theorem 3.8, including both directions of flag transitivity and residual connectedness and the firmness equivalence. For Theorem 3.23 the amalgamation isomorphism must be one fixed map respecting every indexed common parabolic. The common tree theorem and its converse then prove all standard intersections, both directions of flag transitivity, the residue assertions and all panel conditions. This is the coherent interpretation explicitly used on source page 23. The distinctions below explain why the map and the chamber stabilizer cannot be suppressed.

The trivial-Borel condition cannot be omitted. Take both inputs to be the rank-one geometry $(C_2^2, (C_2))$. Each has exactly two vertices and is thin and flag-transitive. Their free-product geometry has two types, and every panel is infinite by the common free-product panel formula. This is a counterexample to the unrestricted thinness assertion in Piedade–Tranchida, arXiv:2505.24662v1, Corollary 3.9. Their definition permits nonfaithful actions; imposing a regular action with trivial Borel recovers the asserted thin case. the common free-product theorem also removes the extra Borel assumption from their converse flag assertion.

1.1 The small free products

Types remain part of a coset system even if they contain one element. Thus Example 3.10's rank-one factor $(1, (1))$ produces the original parabolics together with one equal to the whole group. The result

is the cone on the input, with one new universal element, and is not type-preservingly isomorphic to the input. A rank-zero factor would give the asserted identity. In Example 3.11 the two factors $(C_2, (1))$ give the Bass–Serre tree of $C_2 * C_2$: its vertices are cosets of the two factors, its edges are group elements, and both vertex degrees are two. This is the infinite bivalent tree.

For an n-gon ($n \geq 3$) and the rank-one C_2 factor, the presentation is $\langle a_1, a_2, a_3 \mid a_i^2 = 1, (a_1 a_2)^n = 1\rangle$. The subgroups obtained by deleting one generator are precisely Example 3.12's maximal parabolics. Under the classical Coxeter reflection-triangle realization it is the tessellation with angles $\pi/n, 0, 0$. Amalgamating an n-gon and an m-gon over their shared reflection instead gives

$$\langle a_1, a_2, b_3 \mid a_1^2 = a_2^2 = b_3^2 = 1, (a_1 a_2)^n = (a_2 b_3)^m = 1\rangle.$$

Its maximal parabolics are the three deleted-generator subgroups, as in Example 3.26. For $n, m \geq 3$ the angles $\pi/n, \pi/m, 0$ define a hyperbolic triangle. More generally this interpretation requires $1/n + 1/m < 1$; the algebraic amalgam formula has no such geometric inequality.

Example 3.25's empty shared type set means amalgamation over the identified Borel subgroups, and is a free product if both are trivial. Identifying the two entire systems gives that system. The parenthetical assertion that thin flag-transitive systems always have trivial Borel requires a faithful acting group. It is false for arbitrary specified group actions; the common exact panel formula keeps the Borel explicitly.

1.2 The amalgamation map is part of compatibility

The positive amalgamation theorem uses one chosen isomorphism of the common parabolics carrying every indexed intersection to the matching one. Separate abstract isomorphisms, as written in Definition 3.16, do not imply this coherence for an arbitrary chosen amalgamation. For example take two standard rank-two geometries of C_2^2, with coordinate generators a, b and corresponding maximal subgroups $\langle a\rangle, \langle b\rangle$. Let all types be shared, so the amalgamated subgroup is the whole group, but identify the two copies by the automorphism interchanging a, b. Every pair appearing in Definition 3.16 is abstractly isomorphic. Nevertheless the output maximal subgroup of either type is $\langle a, b\rangle = C_2^2$, since it combines different coordinate lines under the chosen identification. The output consists of two universal singleton types and is not firm, while both inputs are thin.

Thus Lemma 3.17's step from abstract isomorphism to equality of the embedded intersections needs the specified coherent map. The explicit assumption on page 23 that all the maximal parabolics are special supplies exactly that condition. All positive uses here retain it; none applies a gluing theorem to arbitrary abstractly isomorphic parabolic data. This example qualifies the literal shorthand, without refuting the coherent-map theorem proved in the common paper.

1.3 The incompatible symmetric-group parabolics

In Example 3.18 the literal generators a_0, a_1, a_2 satisfy the Coxeter $[4, 3]$ relations and $(a_0 a_1 a_2)^3 = 1$. In the signed permutation model of $[4, 3]$, of order 48, this cube of a Coxeter element is central inversion. Hence $A_3 = \langle a_0, a_1, a_2\rangle$ has order at most 24. The assignment

$$a_0 = (12)(34), \quad a_1 = (23), \quad a_2 = (34), \quad a_3 = (14)(23)$$

satisfies every relator of the full source group in S_4. The first three images generate S_4, proving $A_3 \cong S_4$ exactly. The pair a_0, a_1 generates D_8: its defining order bound is attained in this quotient. Since a_3 commutes with both, $A_2 = \langle a_0, a_1, a_3\rangle$ has order dividing 16. Lagrange's theorem now bounds $|A_2 \cap A_3|$ by $\gcd(16, 24) = 8$, and this intersection contains the displayed D_8. Thus equality holds. On the other side the Coxeter $[3, 3, 4]$ parabolic B_4 is $[3, 3] \cong S_4$ and $B_2 \cap B_4 = \langle b_0, b_1\rangle \cong D_6$. These different intersection orders prove the claimed incompatibility without retaining the classification of the whole regular polytope.

1.4 Why local incidence information is necessary

The following examples concern the unrestricted subgroup statements of N005 Lemmas 3.15(a) and 3.29(a). They do not by themselves disprove the later geometric theorems, whose parabolic hypotheses supply additional structure. In particular, two cosets meeting the same third coset need not meet each other inside that third coset. This is precisely the inference that must be justified when comparing reduced expressions.

Proposition 1. *Special subgroups of a free product with amalgamation need not have the intersection asserted in N005 Lemma 3.15(a).*

Proof. Let

$$A = \langle a, c \mid a^2 = c^2 = [a, c] = 1\rangle, \qquad B = \langle b, c \mid b^2 = c^2 = [b, c] = 1\rangle, \qquad C = \langle c\rangle.$$

Both factors are Klein four groups and $G = A *_C B$ has presentation

$$\langle a, b, c \mid a^2 = b^2 = c^2 = [a, c] = [b, c] = 1\rangle.$$

Set $D_A = \langle a\rangle$, $D_B = \langle b\rangle$, $E_A = \langle ac\rangle$ and $E_B = \langle bc\rangle$. Each of these four groups intersects C trivially. Thus $D = \langle a, b\rangle$ and $E = \langle ac, bc\rangle$ are special subgroups in the exact sense of that lemma, with trivial amalgamated subgroups. Yet

$$ab = (ac)(bc) \in D \cap E.$$

The word ab is nontrivial by the reduced word theorem, whereas $D_A \cap E_A = D_B \cap E_B = 1$. The formula in the lemma would therefore give $D \cap E = 1$, a contradiction.

For completeness, the intersection can be determined exactly. The displayed presentation identifies G with $(C_2 * C_2) \times C_2$. In this product, D is the first factor and E is the graph of the homomorphism $C_2 * C_2 \longrightarrow C_2$ sending both distinguished involutions to the nonidentity element. Its kernel is the infinite cyclic subgroup $\langle ab\rangle$, by alternating reduced words. Hence $D \cap E = \langle ab\rangle$. □

Proposition 2. *Special subgroups of an HNN extension need not have the intersection asserted in N005 Lemma 3.29(a).*

Proof. Use the same Klein four group $A = \langle a, c\rangle$ and identify $C = \langle c\rangle$ with itself by the identity. Its HNN extension is

$$G = \langle a, c, t \mid a^2 = c^2 = [a, c] = [c, t] = 1\rangle.$$

The groups $D = \langle a, t\rangle$ and $E = \langle ac, t\rangle$ are special: both base subgroups meet the associated subgroup C trivially. The asserted intersection formula would give $D \cap E = \langle t\rangle$. However,

$$w = ata = (ac)t(ac) \in D \cap E.$$

To prove $w \notin \langle t\rangle$, map G to $S_3 \times C_2$ by $a \mapsto ((12), 1)$, $t \mapsto ((23), 1)$ and $c \mapsto (1, \epsilon)$, where ϵ generates C_2. Every defining relation is satisfied. The image of w is $((13), 1)$, which is outside the cyclic subgroup generated by $((23), 1)$. □

The same finite group separates the nontrivial word in the first example: send the generators a, b, c to the respective permutations $(12), (23), (45)$ in S_5. The factor groups remain embedded, their intersection is $\langle(45)\rangle$, and ab has order three. These explicit homomorphisms are elementary finite witnesses for the stated nonmembership. No finite quotient is used to assert an equality of subgroups in the infinite groups. The earlier optional GAP verification remains provenance; it is not a proof premise.

The relevant positive property is stronger than arbitrary subgroup compatibility. In a flag-transitive coset geometry, pairwise incident cosets have a common representative. Thus a coset of a parabolic that meets two other incident parabolic cosets meets their intersection. The examples above lack this property for the three local subgroups $C, \langle a\rangle, \langle ac\rangle$. A corrected gluing argument must retain this local incidence information.

2 The exact scope of the HNN construction

The compatible-family HNN theorem in the common paper supplies full coset Helly and all intersections from an indexed partial bijection. Firmness turns the source's unlabelled parabolic-lattice admissibility into precisely that indexed hypothesis. It therefore recovers the whole firm-input branch of Theorem 3.35, with or without residual connectedness. Independently, deleting universal types in a residually connected input reduces its literal source construction to the firm active geometry and then adds back singleton types. Thus the flag-transitivity and residual-connectivity conclusions of part (b) hold even for nonfirm inputs. Stable panels are infinite; the other exact panel indices are the common theorem's indices.

These two positive branches do not imply the unrestricted first sentence or part (a) of Theorem 3.35. We give both the simpler failure of the source intersection formula and a complete counterexample to the broader conclusion. Neither is used as a substitute for the positive arguments.

Why an unlabelled lattice is not sufficient in general

Let A be the additive group of $\mathbf{F}_2^3$, with basis e_1, e_2, e_3, and take four indexed subgroups

$$A_1 = \langle e_2, e_3 \rangle, \quad A_2 = \langle e_1, e_3 \rangle, \quad A_3 = \langle e_1, e_2 \rangle, \quad A_4 = \langle e_3 \rangle = A_1 \cap A_2.$$

This coset geometry is flag-transitive. Indeed, each coset prescribes values of some coordinates. Pairwise intersecting cosets have no conflicting coordinate prescriptions, so all of them have a common representative. Every such flag extends to a chamber by that representative. The geometry is not firm: $H = 0$ and, for example, $R_1 = 0$.

Put $J = K = \varnothing$, so both associated subgroups are A, and let ϕ cyclically permute the basis, $e_1 \mapsto e_2 \mapsto e_3 \mapsto e_1$. The unlabelled standard parabolics are exactly the eight coordinate subspaces. Thus ϕ preserves their whole lattice and is admissible under the literal Definition 3.30. Its action on the three planes is $A_1 \mapsto A_2 \mapsto A_3 \mapsto A_1$. The image $\phi(A_4) = \langle e_1 \rangle$ is a standard parabolic but is not one of the four individually indexed subgroups. Neither is $\phi^{-1}(A_4) = \langle e_2 \rangle$. The equivalence classes in Definition 3.31 are consequently $\{1, 2, 3\}$ and $\{4\}$.

For $s = \{4\}$ the compatibility asserted in the proof of Lemma 3.32 fails:

$$\phi(A_s \cap A) = \langle e_1 \rangle \neq \langle e_3 \rangle = A_s \cap A.$$

The resulting intersection formula on source page 29 is false as well. In the HNN extension

$$G = \langle A, t \mid t^{-1} v t = \phi(v)\ (v \in A) \rangle$$

one has $G_s = \langle t, e_3 \rangle = G$, since conjugating e_3 by t and t^2 supplies e_1 and e_2. Also $F = J \setminus K$ is empty, so $G_t = A_F = A$. It follows that

$$G_t \cap G_s = A, \qquad A_{F \cup s} = A_4 = \langle e_3 \rangle,$$

whose orders are eight and two. Normal forms ensure that A embeds in this HNN extension, so these orders do not collapse.

This example refutes the claimed compatibility and the displayed intersection formula in their stated generality. It does not refute the existence of the abstract HNN extension, nor does it show that G_s cannot be a special subgroup with some larger base subgroup. It also does not disprove the source's broad flag-transitivity conclusion: the particular output just constructed is still flag-transitive. Its maximal parabolics are $\langle t \rangle, G, A$; the G-coset is a single element incident with everything, and $A\langle t \rangle = G$ shows that every pair of the other types meets and has a common representative. The firm case is covered by the common firm-admissibility corollary; more general inputs require explicit indexed compatibility or a separately proved replacement construction.

2.1 The exact source boundary

Throughout, incidence means nonempty intersection of left cosets. The source's Section 2.1, pages 6–8, permits arbitrary indexed subgroups as the standard maximal parabolics; this name does not require maximality in the subgroup lattice. It sets $A_J = \bigcap_{j\in J} A_j$ and $A_\varnothing = A$. A flag-transitive coset incidence system is a geometry, but firmness and residual connectedness are separate conditions.

Definition 3.30, pages 27–28, allows $C = A_J$ and $D = A_K$ for arbitrary $J, K \subseteq I$. An admissible isomorphism $\phi : C \to D$ must preserve the unlabelled lattices of standard parabolics contained in these groups, including their intersections. In particular, $J = K = \varnothing$ and $C = D = A$ are allowed. Theorem 3.35, page 30, assumes only flag transitivity of the input and this admissibility for its initial assertion. Firmness and residual connectedness occur as additional hypotheses for subsequent conclusions. No such hidden assumptions are imposed in the definitions.

For $J = K = \varnothing$, Definition 3.31 identifies named indices whenever $\phi(A_i) = A_j$, taking the equivalence relation generated by these pairs. For a resulting class s it prescribes

$$G_s = \langle t, A_s\rangle, \qquad A_s = \bigcap_{i\in s} A_i, \qquad G_t = A, \qquad G = \langle A, t \mid t^{-1}xt = \phi(x)\ (x \in A)\rangle.$$

Preservation of the unlabelled lattice does not assert that ϕ permutes the individually named A_i. This is the distinction used here.

2.2 An elementary input geometry

In $A = S_{\{1,2,3,4\}}$, put

$$a = (34), \quad b = (12), \quad r = (13)(24), \quad s = (14)(23), \qquad k = ab = rs = (12)(34).$$

Let

$$P = \langle (13), (24)\rangle, \qquad Q = \langle (14), (23)\rangle, \qquad D = \langle a, b, r\rangle,$$

and use the following six indexed subgroups:

$$A_1 = \langle a\rangle, \quad A_2 = \langle b\rangle, \quad A_3 = P, \quad A_4 = Q, \quad A_5 = D, \quad A_6 = \langle r\rangle. \tag{H1}$$

Here D has order eight and preserves the unordered partition $\{\{1,2\},\{3,4\}\}$. Its normal forms are $a^i b^j r^\ell$, with $i, j, \ell \in \{0,1\}$; conjugation by r swaps a and b. The standard parabolics are exactly

$$\mathcal{L} = \{A, D, P, Q, \langle a\rangle, \langle b\rangle, \langle r\rangle, \langle s\rangle, 1\}. \tag{H2}$$

Indeed, $D \cap P = \langle r\rangle$, $D \cap Q = \langle s\rangle$, and $P \cap Q = 1$. All four listed order-two groups lie in D; among them only $\langle r\rangle$ lies in P and only $\langle s\rangle$ lies in Q. Distinct order-two groups intersect trivially. These observations prove intersection closure and show that every member of (H2) is a standard parabolic of (H1).

Lemma 3. *Every pairwise-intersecting family of left cosets of members of $\mathcal{L}$ has a common representative.*

Proof. Cosets of A impose no restriction. A coset of 1 is a singleton, so pairwise intersection with it already gives the conclusion. Repeated cosets can be removed; distinct cosets of the same subgroup are disjoint. It remains to consider at most one coset of each of the other seven groups.

Call cosets of $\langle a\rangle, \langle b\rangle, \langle r\rangle, \langle s\rangle$ atomic. These are edges with endpoints x, xz, where $z \in \{a, b, r, s\}$. Every atomic coset lies in one left coset of D. Pairwise-intersecting atomic cosets therefore lie in the same D-coset. The homomorphism

$$\epsilon : D \longrightarrow C_2, \qquad \epsilon(a^i b^j r^\ell) = i + j + \ell \pmod 2$$

takes all four of $a, b, r, s = abr$ to 1. Thus their Cayley graph on each D-coset is bipartite. Pairwise-intersecting edges in a triangle-free simple graph have a common vertex: otherwise two edges meeting at x would be met away from x by a third edge, giving a triangle.

Suppose at least two atomic cosets occur, with common vertex x. Their intersection is exactly $\{x\}$. A D-coset meeting an atomic coset contains it, and hence contains x. Consider a P-coset meeting all the atomic cosets. If the $\langle r\rangle$-coset occurs, it is contained in that P-coset, so again x belongs. Otherwise take two distinct atomic generators from a, b, s. If the P-coset omitted x, it would contain both opposite endpoints xz_1, xz_2, forcing $z_1 z_2 \in P$. The product $ab = k$ is not in P, and as, bs have order four, whereas P has exponent two. This is impossible. For Q the same argument uses s in place of r and the three remaining generators a, b, r. Thus x lies in every coset in the family.

Suppose exactly one atomic coset occurs. A D-coset is then redundant. For the $\langle r\rangle$-coset a P-constraint is redundant, and for the $\langle s\rangle$-coset a Q-constraint is redundant; the remaining pairwise constraints have a common point. For an atom of type a or b, the only remaining possible failure would have a P-coset meeting one endpoint and a Q-coset meeting the other. Their mutual intersection would imply $z \in PQ$ (or QP) for $z = a$ or b. This does not happen. Multiplying a or b by the four members $1, (14), (23), s$ of Q gives, respectively, the transposition a or b, two three-cycles, and a four-cycle, none in P. Inversion gives the same exclusion for QP.

Finally suppose no atomic coset occurs. Only a P-, a Q-, and a D-coset remain. If all three occur, translate the unique intersection of the first two to 1, so those two cosets are P, Q. Write the third as gD. It meets both only if $g^{-1} \in DP \cap DQ$. But

$$DP = D \cup D(13), \qquad DQ = D \cup D(14), \qquad DP \cap DQ = D.$$

The two nontrivial right D-cosets are distinct, since $(13)(14)$ has order three and cannot lie in the group D of order eight. Hence $g \in D$, so the third coset contains 1 as well. □

Lemma 3 proves that every flag for (H1) has a common representative x. It is therefore a translate by x of a standard flag and extends to $\{xA_i : 1 \le i \le 6\}$. Thus the input is a flag-transitive coset geometry, with no reliance on the finite enumeration. Its Borel is 1. Every minimal parabolic is 1, since at least two of the three distinct groups $\langle a\rangle, \langle b\rangle, \langle r\rangle$ remain when any single type is deleted. Every standard panel consequently has one chamber, so the geometry is not firm. It is not residually connected either: the residue of the standard flag of types $\{1, 5\}$ has group $\langle a\rangle$ and four remaining subgroups all equal to 1. Its incidence graph is two disjoint complete graphs on four vertices.

2.3 Admissibility and the literal output

Define $\phi(x) = bxb$. It fixes $A, D, \langle a\rangle, \langle b\rangle, 1$ and interchanges P, Q and $\langle r\rangle, \langle s\rangle$. Hence it preserves exactly the lattice (H2) and all intersections: it satisfies Definition 3.30. The named index classes in Definition 3.31 are

$$\{1\}, \quad \{2\}, \quad \{3, 4\}, \quad \{5\}, \quad \{6\}. \tag{H3}$$

In particular, $\phi(A_6) = \langle s\rangle$ is a standard parabolic but is not one of the six named A_i.

For clarity, use the following coordinate convention for the infinite group. Represent xt^n by $(x, n) \in A \times \mathbb{Z}$ and put

$$(x, n)(y, m) = (x\phi^{-n}(y), n + m). \tag{H4}$$

Associativity follows from the automorphism law, and $(1, -1)(x, 0)(1, 1) = (\phi(x), 0)$. Conversely the relation in the HNN presentation moves every letter from A to the left, giving xt^n; the map into (H4) shows uniqueness. Thus (H4) is precisely the presented HNN extension, not merely a quotient of it. In this example $\phi^{-n} = \phi^n$ because $\phi^2 = 1$.

Let $V = \langle r, s\rangle = \{1, r, s, k\}$. For each ϕ-invariant subgroup $B \le A$, the subgroup $\langle B, t\rangle$ is the set $B \times \mathbb{Z}$ under (H4). The literal maximal parabolics of the output are therefore

type	subgroup in coordinates (H4)
$\{1\}$	$\langle a\rangle \times \mathbb{Z}$
$\{2\}$	$\langle b\rangle \times \mathbb{Z}$
$\{3, 4\}$	$\{1\} \times \mathbb{Z}$
$\{5\}$	$D \times \mathbb{Z}$
$\{6\}$	$V \times \mathbb{Z}$
t	$A \times \{0\}$

(H5)

Here the sixth input group expands because conjugation by t changes r to s, and $P \cap Q = 1$ accounts for the third row.

Theorem 4. *The output of Definition 3.31 for (H1) and $\phi(x) = bxb$ is neither flag-transitive nor a geometry.*

Proof. Write G_1, G_2, G_6 for the output groups at the corresponding singleton classes. The three distinct-type cosets

$$G_1, \qquad G_2, \qquad aG_6 \tag{H6}$$

meet pairwise: their respective intersections contain $1, a, b$, since $ab = k \in V$. On the other hand,

$$G_1 \cap G_2 = \{1\} \times \mathbb{Z} = \langle t\rangle, \qquad aG_6 = aV \times \mathbb{Z}.$$

The coset aV omits 1 (its elements are odd permutations), so (H6) has empty total intersection. A translate of the standard flag has a common representative. Hence this flag cannot be in its orbit, proving failure of flag transitivity.

More strongly, the output type $\{3, 4\}$ has subgroup $\langle t\rangle$. Any one of its cosets incident with both G_1 and G_2 is contained in both: $\langle t\rangle$ is a subgroup of each. Such a coset must be exactly $\langle t\rangle$, which does not meet aG_6. Thus (H6) cannot extend to a chamber. Since the type set is finite, it extends to a maximal flag that is not a chamber. The output is not a geometry. □

2.4 The simplex HNN construction

Take $a_i = (i, i+1)$ in S_6, $1 \le i \le 5$, as in Example 3.36. The associated subgroups are $C = \langle a_2, a_3\rangle$ and $D = \langle a_3, a_4\rangle$, with indexed map $a_2 \mapsto a_3$, $a_3 \mapsto a_4$. The type classes are $\{1\}, \{2, 3, 4\}, \{5\}$ and $F = \{4\}$. The indexed HNN theorem in the common paper therefore gives

$$G_1 = \langle A_1, t\rangle, \quad G_{234} = \langle a_1, a_5, t\rangle, \quad G_5 = \langle A_5, t\rangle, \quad G_t = A_4 = \langle a_1, a_2, a_3, a_5\rangle.$$

Indexed compatibility follows from the named bijection and classical Coxeter parabolic embeddings. The resulting geometry is flag-transitive, residually connected and firm. Its Borel is trivial, and the exact panel formula gives $2, 6, 2, \infty$. Example 3.37's whole-base automorphism gives $A \rtimes \mathbb{Z}$ with the stated orbit-coarsened parabolics when its indexed action is specified. For firm inputs the Boolean-parabolic lemma in the common paper supplies the indexed action from source admissibility. Preservation of an unlabelled parabolic family alone does not supply it on an arbitrary nonfirm input.

3 Two semidirect constructions

The first construction, in source Section 4.1, is the join described in the common paper. It is flag-transitive and has parabolics $A_E \rtimes B_J$; its residual connectedness and firmness follow there. For thin inputs it is thin exactly when every old type orbit is a singleton. Consequently the literal claim that it is never thin is too strong. In particular, trivial type action gives the join of the original thin geometries. No faithfulness or normality of a Borel subgroup was used in the exact panel argument.

The second construction has different parabolics. Its identification with the common orbit construction is given next, including the thinness qualification that survives nonfaithful actions.

3.1 Literal subgroups and the common orbit theorem

Write $\alpha = (A, (A_x)_{x\in S})$ and $\beta = (B, (B_i)_{i\in I})$ for flag-transitive finite-rank coset geometries, with $A_E = \bigcap_{x\in E} A_x$ and $B_J = \bigcap_{i\in J} B_i$. Empty intersections are the ambient groups. Fix an indexed action of B on S together with an action on A satisfying $b(A_x) = A_{bx}$. This indexed action is necessary data if distinct types have identical maximal parabolics; permutation of the unlabelled subgroup set does not itself specify it.

In source Definition 4.5 there is one nonsingleton orbit L, a representative $x_0 \in L$, and singleton orbits elsewhere. Translate its IPO convention into

$$B_J x_0 \cap B_K x_0 = B_{J\cup K} x_0 \qquad (J, K \subseteq I).$$

The same equation holds automatically on a singleton orbit. Choose its unique element as representative. Let $\mathcal{K}$ be all these orbits and set $T = I \sqcup \mathcal{K}$. For $J \subseteq T$ put

$$J_B = J \cap I, \quad J_A = J \cap \mathcal{K}, \qquad E(J) = S \setminus \bigcup_{M\in\mathcal{K}\setminus J_A} B_{J_B} x_M.$$

The common orbit theorem gives the exact standard intersections

$$G_J = A_{E(J)} \rtimes B_{J_B}, \qquad G_T = A_S \rtimes B_I, \quad G = A \rtimes B. \tag{1}$$

For a singleton orbit $M = \{y\}$, the union includes y unless that orbit type belongs to J_A. Consequently $E(J) = (L \setminus B_{J_B} x_0) \cup \bigcup_{M\in J_A} M$. At a maximal parabolic this is exactly Definition 4.6. Thus no generated-support identity, or an interchange between quotient images and intersections, is part of this identification.

The same theorem proves flag transitivity and chamber completion, finiteness if and only if both inputs are finite, and preservation of firmness and residual connectedness under their stated input hypotheses. In particular it recovers Lemmas 4.7–4.8, Propositions 4.9–4.11 and Theorem 4.12(a)–(d), with the corrected B_{J_B} notation in the second factor of Lemma 4.8. Its exact panel indices are

$$[G_{T\setminus\{i\}} : G_T] = [B_{I\setminus\{i\}} : B_I] \qquad (i \in I),$$
$$[G_{T\setminus\{M\}} : G_T] = [A_{S\setminus B_I x_M} : A_S] \qquad (M \in \mathcal{K}).$$

These recover Theorem 4.12(e)'s firmness assertion. If the inputs are thin and B_I fixes every x_M, every displayed index is two. In particular faithful regular input actions give the regular conclusion of Corollary 4.13, with faithful full type-preserving group $A \rtimes B$ when both chamber stabilizers are trivial. The common theorem's chamber count is $[G : G_T] = [A : A_S][B : B_I]$, including infinite cardinalities; finiteness refers to the geometries, not to their possibly unfaithful acting groups.

3.2 Why thinness needs the chamber-stabilizer condition

Here is the literal obstruction when thin coset models may act unfaithfully. It is the same complete coordinate example already used for the later source on wreath products; we give its N005 adapter explicitly. Take $A = C_2^3$ with the three coordinate hyperplanes as its maximal parabolics. Its coset geometry is the join of three two-point sets, a faithful regular rank-three geometry. Let $B = S_3$ permute the coordinates and let $\beta = (S_3, (C_3))$. This is a thin rank-one geometry with two elements and nontrivial chamber stabilizer C_3.

There is one orbit L of size three. Both B and C_3 are transitive on it, so the entire IPO condition holds, including the empty and full subsets of the one-element type set. The source twisting subgroups are, literally,

$$P_L = 1 \rtimes S_3, \qquad P_i = C_2^3 \rtimes C_3.$$

Their intersection is $1 \rtimes C_3$. Every coset of P_i meets every coset of P_L: $P_i P_L = G$ because their factors contain the whole base and the whole acting group. Hence the geometry is the complete bipartite graph

on parts of sizes two and eight. Its two panel sizes are eight and two; it is firm, residually connected and flag-transitive, but is not thin. Every assertion here follows directly from the displayed finite groups, without a computation.

The source's proof of Theorem 4.12(e) replaces thinness by triviality of both Borel subgroups. That implication requires faithfulness and is false for the coset model above. Its thinness assertion and the unqualified regular-input reading of Corollary 4.13 must therefore retain the stated action hypothesis or the exact panel criterion. This is a correction to literal scope, not a failure of the faithful regular twisting theorem. No numbered-conjecture or historical priority claim is attached to it.

4 Artin and Shephard applications

The common cyclic-marking theorem proves all intersection, coset Helly, residue and panel assertions after an exact semidirect-product identification. Its two presentation theorems supply inverse maps, rather than only representations of a quotient. We retain the classical standard-parabolic inputs for Artin and Coxeter groups, with their exact references stated in the common paper. General Shephard groups do not receive those stronger properties merely from their presentations. The following separate argument supplies exactly the generator and diagram-action statement needed for source Lemma 5.6.

4.1 Separating the named Shephard generators

A properly labelled Shephard graph has a finite vertex set S, vertex orders $p_i \in \{2, 3, \ldots\} \cup \{\infty\}$, and pair labels $m_{ij} \in \{2, 3, \ldots\} \cup \{\infty\}$. An odd finite pair label requires $p_i = p_j$. Its group has presentation

$$\langle s_i \ (i \in S) \mid s_i^{p_i} = 1 \ (p_i < \infty), \quad \mathsf{W}_{m_{ij}}(s_i, s_j) = \mathsf{W}_{m_{ij}}(s_j, s_i) \ (m_{ij} < \infty) \rangle,$$

where W_m denotes an alternating word of length m. A pair of label two commutes, while an infinite label imposes no pair relation. All vertex orders infinite give the Artin presentation; all vertex orders two give the Coxeter presentation. Graph automorphisms preserve both sets of labels. We now close the injectivity point in source Lemma 5.6 without assuming an embedding theorem for general Shephard groups. This argument constructs a representation separating the named generators; it does not assert that the representation is faithful on the entire group.

Proposition 5. *In the presentation of Definition 5.1, every named generator has its prescribed order and distinct names give distinct elements. Consequently the action of the labelled graph automorphism group on the presented Shephard group is faithful.*

Proof. Let V have basis $(e_i)_{i \in S}$ over $\mathbb{C}$. For vertex order $p_i < \infty$ choose $z_i = \exp(2\pi \mathrm{i}/p_i)$, and for infinite order choose $z_i = 2$. Thus $z_i \neq 1$ in every case, and equal vertex orders give equal z_i. Fix square roots $u_i^2 = z_i$. Define linear maps

$$R_i(e_i) = z_i e_i, \qquad R_i(e_j) = e_j + c_{ij} e_i \quad (j \neq i).$$

Choose the two coefficients independently for each unordered pair, after ordering its endpoints as $i < j$. For a commuting pair take $c_{ij} = c_{ji} = 0$; for label infinity do the same, since no braid needs to be imposed. For finite labels greater than two take

$$c_{ij} = 1, \qquad c_{ji} = \begin{cases} -4z_i \cos^2(\pi/m), & m \text{ odd}, \\ 2u_i u_j \cos(\pi/k) - z_i - z_j, & m = 2k. \end{cases}$$

Proper labelling ensures $z_i = z_j$ in the odd case. The maps are invertible, with determinant z_i. The geometric-series formula for their exceptional row shows $R_i^{p_i} = 1$ for finite p_i; its eigenvalue z_i gives exact order p_i. If $p_i = \infty$, the eigenvalue two gives infinite order.

We verify the braid relations on the whole vector space. On $V_{ij} = \langle e_i, e_j \rangle$, put

$$A = \begin{pmatrix} z & c \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ d & w \end{pmatrix}, \qquad P = AB.$$

For $m = 2k$, the trace and determinant of P are $2uv\cos(\pi/k)$ and $(uv)^2$, where $z = u^2, w = v^2$. Its distinct eigenvalues are $uv\exp(\pm\pi\mathrm{i}/k)$. Thus $P^k = -(uv)^k I = (BA)^k$, the required braid.

For $m = 2r + 1$, put $z = w$, $q = -\cos(2\pi/m)$ and use $cd = -4z\cos^2(\pi/m)$. Then $\operatorname{tr} P = 2zq$ and $\det P = z^2$. Let $U_{-1} = 0$, $U_0 = 1$ and $U_{j+1}(q) = 2qU_j(q) - U_{j-1}(q)$. Cayley–Hamilton gives

$$P^r = z^{r-1}U_{r-1}(q)P - z^r U_{r-2}(q)I.$$

Direct multiplication also gives $PA - BP = (z + cd)(A - B)$. Since $1 + cd/z = 2q - 1$, subtraction yields

$$P^r A - BP^r = z^r\big(U_r(q) - U_{r-1}(q)\big)(A - B) = 0.$$

For the last equality write $q = \cos\theta$ with $\theta = \pi - 2\pi/m$. The recurrence has solution $U_j(q) = \sin((j+1)\theta)/\sin\theta$; hence the difference is zero because $\cos(m\theta/2) = 0$. The two sides before subtraction are precisely the alternating words of length $2r + 1$.

To pass from V_{ij} to V, set $D = (z - 1)(w - 1) - cd$. When $D \neq 0$, the common fixed space of R_i, R_j is complementary to V_{ij}: solve its two fixed-vector linear equations by adjusting the e_i, e_j coordinates. Both maps are the identity on that complement, so the verified pair relation holds on V. The exceptional values $D = 0$ cause no gap. In the odd case vary z while keeping all other row coefficients fixed and $cd = -4z\cos^2(\pi/m)$; D is a nonzero polynomial in z. Every entry of the braid difference is a polynomial vanishing at all but finitely many complex z, and thus vanishes also at the exceptional values. In the even case vary u with the nonzero v fixed; $D = u^2v^2 + 1 - 2uv\cos(\pi/k)$ is again a nonzero polynomial. The same polynomial argument applies, including the finitely many excluded zero values. For a commuting pair the off-diagonal pair coefficients are zero and the two exceptional rows commute directly. All defining relations therefore hold on V.

The presentation now gives a homomorphism to $\mathrm{GL}(V)$. If $i \neq j$, the ith diagonal entry of R_i is $z_i \neq 1$, whereas that of R_j is one. Thus these matrices, and consequently the named group elements, are distinct. A labelled graph permutation and its inverse preserve all defining relators, so they induce inverse group automorphisms. A nonidentity graph permutation moves a named generator to a different one, proving faithfulness. □

The constructed representation is used only to separate names and certify their orders. No parabolic embedding, intersection or coset Helly property for arbitrary Shephard groups follows from it.

4.2 Free products and quotient actions

If A, B are Artin or Coxeter groups with disjoint standard markings, $A * B$ has the union presentation, equivalently all cross labels are infinite. Its generated subgroup on R is $A(R \cap S_A) * B(R \cap S_B)$ by reduced normal forms. Two such subgroups intersect by intersecting the two factor markings: the unique alternating reduced word belongs to both exactly when each syllable does. The coset Helly theorem for free products proved in the common paper then applies to the two classical input geometries. Its hypotheses allow arbitrary Borel subgroups; here both Borels are trivial. The resulting standard marking therefore satisfies the common parabolic-marking lemma. This recovers source Theorem 5.4 and all three clauses of Corollary 5.5, for arbitrary choices of the two Artin/Coxeter factors. Each source panel size is preserved.

4.3 The involutory and cyclic families

Apply the common involutory presentation theorem to two leaves with equal labels to every remaining vertex. This gives source Theorems 5.14–5.15 for every allowed finite or infinite label, with all their parabolic product and flag-complex conclusions. Its special D_n case fixes the diagram convention explicitly:

For type D_n ($n \geq 4$), take the chain $a_1, a_2, \dots, a_{n-1}$ with consecutive label three, attach a_n by label three to a_2, and let all other distinct pairs commute. Interchange a_1, a_n. The new marking $t, a_1, a_2, \dots, a_{n-1}$ has vertex orders $2, \infty, \dots, \infty$, consecutive labels $4, 3, \dots, 3$ and all nonconsecutive commutations. It is exactly $B_n(2, \infty)$. This proves source Propositions 5.12–5.13; the arbitrary labels in the common involutory theorem prove Theorems 5.14–5.15, including every stated intersection and flag-complex assertion. The t panels have size two and all remaining panels are infinite.

There are two literal presentation qualifications. On source page 46 the graph attaches a_n to a_2, whereas the printed relator list uses a_3 and does not consistently bound the chain index. The proof on page 47 uses the graph's a_2 convention, as do we. Also the $B_n(p, q)$ display on page 41 omits the nonconsecutive commutations; they are required by Definition 5.1 and by the displayed graph. They are explicitly retained here. These are corrected graph presentations, not verification of the malformed relator lists as printed.

For cyclic amalgamation take k copies of the Artin group on $Y \sqcup \{x\}$ over the full standard parabolic on Y. The universal property identifies the amalgam M with the Artin presentation having k leaves and no relations between distinct leaves. The common cyclic presentation theorem gives exactly $M \rtimes C_k$, with vertex t of order k, no relation on x_0, t, and all mixed braids inherited from one copy. Its complete inverse maps and cyclic-marking proof supply every geometric conclusion, with panel sizes k and infinity.

These are source Theorems 5.16–5.17. The source's literal statement writes $A_0 \rtimes C_k$ where its construction and proof require $M \rtimes C_k$. There need not even be the indicated cyclic action on the single copy A_0. Its page 51 preliminary braid display also has identical words on both sides; the right side must start with y. When $k = 2$, the order-two vertex is represented by no loop under Definition 5.1, rather than by a loop labelled two. None of these conventions affects the corrected presentation just proved.

4.4 The tetrahedral Artin example

In Example 5.7 let A be the Artin group on a_1, a_2, a_3, a_4 with all pair labels three. The permutations $p = (23)$ and $q = (34)$ generate a group B acting on the names. At representative 2 the orbits of $1, \langle q \rangle, \langle p \rangle, B$ are $\{2\}, \{2\}, \{2, 3\}, \{2, 3, 4\}$. These give the exact orbit-intersection condition; the fixed singleton orbit $\{1\}$ also satisfies it. The complete orbit-transfer theorem applies. The maximal parabolics are

$$A(\{2,3,4\}) \rtimes B, \quad A(\{1\}) \rtimes B, \quad A(\{1,2\}) \rtimes \langle q \rangle, \quad A(\{1,2,3\}) \rtimes \langle p \rangle.$$

They are the deleted-generator subgroups of (a_1, a_2, p, q). This proves the source's firm, residually connected, flag-transitive rank-four geometry; its panels have sizes $\infty, \infty, 2, 2$.

There is also a four-generator presentation, with $a = a_1, x = a_2$:

$$p^2 = q^2 = (pq)^3 = 1, \qquad [a, p] = [a, q] = [x, q] = 1,$$
$$\mathsf{W}_3(a, x) = \mathsf{W}_3(x, a), \qquad \mathsf{W}_6(x, p) = \mathsf{W}_6(p, x).$$

Set $x_2 = x$, $x_3 = pxp$, $x_4 = qpxpq$. The relation $pqp = qpq$ and $[x, q] = 1$ show that p swaps x_2, x_3 and fixes x_4, while q swaps x_3, x_4 and fixes x_2. The doubled-braid identity converts the length-six relation into the length-three braid on x_2, x_3. Conjugation supplies the other pair braids and the braids with a. These formulas give mutually inverse maps with the original six-generator semidirect presentation.

5 The affine quotient and its literal twist

The input geometry and its twist must be proved separately. The input has the required residual geometry for every $s \geq 2$, using a uniform coordinate argument for $s \geq 5$ and the complete finite certificate for $s = 2, 3, 4$. Its failure at $s = 1$ does not prevent the literal twist from being regular. The latter conclusion requires actual intersections of quotient maximal parabolics, not an assumed substitution by images of generated supports. Both bridges are included below.

The following argument proves the positive input assertion in source Example 5.11. It does not retain the cited Ens classification as a proof premise. It is separate from the written correction of the source's assertion about the smallest twist.

Retain the classical faithful affine Coxeter realization $W = Q \rtimes S_4$, where $Q = \{v \in \mathbb{Z}^4 : \sum v_i = 0\}$, with multiplication $(\pi, v)(\rho, w) = (\pi\rho, v + \pi w)$. The semidirect model is the standard affine Weyl description; an explicit reference is P. Etingof–X. Ma, *Lecture notes on Cherednik algebras*, §7.12, p. 58. Faithfulness for the displayed affine Coxeter presentation remains the retained classical affine reflection theorem; the coordinate calculations below do not prove that external theorem anew. Take the literal four reflections

$$a_1 = ((12), 0), \quad a_2 = ((23), 0), \quad a_3 = ((14), e_1 - e_4), \quad a_4 = ((34), 0).$$

They realize the square diagram with the commutations on pairs 14 and 23. Put $w = a_3a_4a_1a_2$. Then $w^2 = (1, (1, 1, -1, -1))$, and its $2s$-power relation has normal closure sM, where

$$M = 2Q + \mathbb{Z}(1, 1, -1, -1) = \{v \in Q : v_i \equiv v_j \pmod 2 \text{ for all } i, j\}.$$

Indeed its coordinate-permutation orbit generates exactly this lattice: differences of orbit vectors give $2(e_i - e_j)$, and all orbit vectors agree modulo $2Q$. Translations commute, so these conjugates are the entire normal closure. Thus the literal quotient is $W_s = (Q/sM) \rtimes S_4$ and has order $96s^3$.

Theorem 6. *For every integer $s \geq 2$, the standard four-generator coset system of W_s is a faithful regular hypertope. The cases $s = 2, 3, 4$ use the explicit finite certificate below; all $s \geq 5$ follow from the uniform argument in this proof. The case $s = 1$ fails residual connectedness, as proved in the literal-twist argument below.*

Proof. The maximal proper standard parabolics of W fix, respectively,

$$c_1 = (3, -1, -1, -1)/4, \quad c_2 = (1, 1, -1, -1)/2, \quad c_3 = 0, \quad c_4 = (1, 1, 1, -3)/4.$$

Each has linear part S_4, because its three generating transpositions form a spanning tree on the four coordinates. An element fixing c_i has translation $c_i - \pi c_i$. Its translation coordinates therefore lie in $\{-1, 0, 1\}$ and its order is 24. Every proper generated parabolic is contained in one of these four. A product of two proper parabolic elements has all translation coordinates in $[-2, 2]$.

For $s \geq 5$, two such products with the same quotient image have equal permutation parts and translation difference in sM. Every coordinate of that difference has absolute value at most four, and is a multiple of s; hence it is zero. The quotient is therefore injective on the union of all products of two proper parabolics. The classical Coxeter identities

$$W(R) \cap W(S) = W(R \cap S), \qquad W(R)W(S) \cap W(R)W(T) = W(R)W(S \cap T)$$

descend to the quotient: lift an element of the left side to the two indicated products; injectivity makes the lifts equal and the identity in W supplies the required right-side representative. The identity subgroup is a proper parabolic, so this also proves the first formula. Any occurrence of the whole group is trivial. The images of all generated parabolics are consequently the actual intersections of the maximal parabolics in the quotient.

The product identity implies coset Helly for this intersection-closed family. Three pairwise-intersecting cosets can be translated into H, zK, zL; the identity $HK \cap HL = H(K \cap L)$ supplies a common element. Replace two cosets by their intersection coset and induct to obtain the finite Helly property. Thus every flag extends and is a translate of a standard flag. Each standard residue is a generated parabolic with its children obtained by deleting one remaining name. If at least two remain these children generate it, so its incidence graph is connected. This proves residual connectedness. The Borel is trivial by the intersection formula and all singleton parabolics have order two, proving thinness. A chamber-fixing automorphism fixes all neighboring chambers and hence all chambers, so the regular action is the full type-preserving one.

For $s = 2, 3, 4$, the complete exact certificate below supplies the same intersection and product identities and nontrivial singleton orders. The identical coset argument gives the result in these remaining cases. □

The required finite certificate

The complete file `programs/affine_example.py` uses only integer arithmetic, four-coordinate permutations and finite sets. It enumerates every proper subgroup generated by a subset of the four literal affine reflections. Breadth-first multiplication is continued until the set closes; no expected order is used to stop the enumeration. A cap rejects an unexpectedly infinite computation. For binary masks 0 through 14 the resulting orders are

$$1, 2, 2, 6, 2, 6, 4, 24, 2, 4, 6, 24, 6, 24, 24.$$

These sets are exactly the generated subgroups, by closure and reachability of every enumerated element. Reduction modulo sM first reduces the four coordinates modulo $2s$, and then identifies a vector with its translate by (s, s, s, s). The formula for M proves that this is an exact normal-form equality test, not a separating quotient approximation.

For each of $s = 2, 3, 4$, the program verifies that each proper parabolic retains its size, all $15^2 = 225$ pair intersections equal the expected subgroup, and all $15^3 = 3375$ product identities hold. Thus all three positive cases are fully tested, including the trivial subgroup; whole-group cases are tautologies as in the proof. The program also checks the affine Coxeter relations and the displayed translation word. Its additional quotient-order and $s = 1$ residue checks validate the written calculation and are not used to extrapolate the positive range. The complete program, its interpreter semantics and the retained faithful affine realization are explicit proof inputs. No certificate or finite sample substitutes for the $s \geq 5$ argument. The bound here is on individual coordinates, not an incorrect bound of two on the translation ℓ^1 norm of a proper parabolic.

5.1 The source construction and its coordinates

Let W have the literal presentation on source page 45:

$$W = \langle a_1, a_2, a_3, a_4 \mid a_i^2 = 1,\ [a_1, a_4] = [a_2, a_3] = 1, (a_1a_2)^3 = (a_1a_3)^3 = (a_2a_4)^3 = (a_3a_4)^3 = 1 \rangle.$$

Set $w = a_3a_4a_1a_2$, $N_s = \langle\langle w^{2s} \rangle\rangle$ and $A_s = W/N_s$, for $s \geq 1$. Let p and q permute the four named generators by (12)(34) and (23), respectively. They preserve the defining Coxeter relations. Moreover,

$$p(w) = (a_4a_3)w^{-1}(a_3a_4), \qquad q(w) = (a_3a_2)w(a_2a_3). \tag{2}$$

The first identity is cancellation; the second also uses $a_2a_3 = a_3a_2$. Hence they preserve N_s and induce an action of the abstract diagram group $B = \langle p, q \mid p^2 = q^2 = (pq)^4 = 1 \rangle \cong D_8$ on A_s. The inverse permutations ensure equality of the normal closures, rather than just one inclusion. Put $G_s = A_s \rtimes B$. We use left-action coordinates $(a, b)(c, d) = (a\, b(c), bd)$, so $bab^{-1} = b(a)$. For the involutions p, q this agrees with the source's conjugation notation without a choice of inverse action.

In the source take $B_p = \langle q \rangle$, $B_q = \langle p \rangle$, and choose $F_0 = 1$. The orbits of 1 under $1, B_p, B_q, B$ are, respectively, $\{1\}, \{1\}, \{1,2\}, \{1,2,3,4\}$. Thus its orbit-intersection condition holds. The following are the literal three maximal parabolics of the twisting system, as proved in Lemma 7 rather than inferred from a quotient intersection property:

$$P_L = B, \qquad P_p = \langle a_1 \rangle \rtimes \langle q \rangle, \qquad P_q = \langle a_1, a_2 \rangle \rtimes \langle p \rangle. \tag{3}$$

In particular these subgroups can be studied even when the input coset system fails residual connectedness. We do not invoke a transfer theorem whose input hypotheses are then absent.

5.2 Identifying the literal quotient parabolics

We retain the standard faithful affine Coxeter realization of the literal presentation. This external input is needed here to identify the actual quotient intersections in the source construction, as well as for the order calculation and input-failure discussion below. Let $Q = \{v \in \mathbb{Z}^4 : \sum v_i = 0\}$ and let (π, v) act by $z \mapsto \pi z + v$. In $W = Q \rtimes S_4$ take

$$a_1 = ((12), 0),\ a_2 = ((23), 0),\ a_3 = ((14), e_1 - e_4),\ a_4 = ((34), 0).$$

Direct multiplication gives $w^2 = (1, (1, 1, -1, -1))$. Consequently N_s is the lattice sM, where

$$M = 2Q + \mathbb{Z}(1, 1, -1, -1) = \{v \in Q : v_i \equiv v_j \pmod 2 \text{ for every } i, j\}.$$

Indeed conjugation permutes the coordinates of the translation; differences of orbit vectors generate $2Q$, and every orbit vector is congruent to $(1, 1, -1, -1)$ modulo $2Q$. Thus $[Q : M] = 4$, $|A_s| = 96s^3$ and $|G_s| = 768s^3$. The affine model is an explicit retained premise of the literal-source identification. The subsequent abstract marking argument uses only the source presentation and an elementary S_4 quotient.

Lemma 7. *For every $s \geq 1$, the actual quotient maximal parabolics satisfy*

$$(A_s)_3 \cap (A_s)_4 = \langle a_1, a_2 \rangle, \quad (A_s)_2 \cap (A_s)_3 \cap (A_s)_4 = \langle a_1 \rangle, \quad \bigcap_{i=1}^{4} (A_s)_i = 1.$$

Consequently the literal twisting subgroups are exactly (3).

Proof. The four maximal parabolics of W fix the points

$$c_1 = (3, -1, -1, -1)/4, \quad c_2 = (1, 1, -1, -1)/2, \quad c_3 = 0, \quad c_4 = (1, 1, 1, -3)/4.$$

Each has linear part S_4: its three generating transpositions form an edge tree on the four coordinates. An element fixing c_i has translation $c_i - \pi c_i$. The quotient embeds each such parabolic, since a pure translation fixing a point is the identity.

The third parabolic is the pure linear S_4. Its element π is in the image of the fourth precisely when $c_4 - \pi c_4 \in sM$. A nonzero such difference is a vector with exactly two nonzero coordinates, one 1 and one -1. It cannot belong to sM: for $s \geq 2$ divisibility fails, and for $s = 1$ its four coordinate parities are not equal. Thus π must fix coordinate 4. These elements form exactly $\langle (12), (23) \rangle = \langle a_1, a_2 \rangle$.

Within this subgroup, membership in the second parabolic means $c_2 - \pi c_2 \in sM$. Such a difference has all coordinates in $\{-1, 0, 1\}$. It is zero exactly when π preserves both blocks $\{1, 2\}, \{3, 4\}$. The only other way to have all equal coordinate parities is to interchange the blocks, which is impossible when coordinate 4 is fixed. Thus π preserves the blocks and fixes both 3 and 4, leaving only $\langle (12) \rangle$. Finally membership in the first parabolic forces coordinate 1 to be fixed, by the same two-odd-coordinates argument for $c_1 - \pi c_1$. This leaves the identity, proving all three formulas. □

5.3 A uniform direct proof of regularity

Theorem 8. *For every $s \geq 1$, the literal coset system $(G_s, (P_L, P_p, P_q))$ is a thin, residually connected, flag-transitive rank-three geometry. Its distinguished marking is (x, p, q) with $x = a_1$, and its adjacent product orders are six and four. In particular it is a regular hypertope at $s = 1$.*

Proof. The action of B on the four names is transitive. Therefore x, p, q generate G_s. The pair a_1, a_2 generates an exact S_3 in every A_s. Its order is at most six by its two involutions and braid of length three. For the lower bound use the quotient map

$$a_1 \mapsto (12), \quad a_2 \mapsto (23), \quad a_3 \mapsto (14), \quad a_4 \mapsto (34) \quad \text{in } S_4.$$

All defining relations hold and the image of w is $(12)(34)$, so the map kills w^{2s} for every s. The first two images generate S_3, proving the lower bound and the claimed injection. The same S_4 image distinguishes all four named generators in A_s. The diagram group D_8 permutes these names faithfully, so its action on A_s is faithful as well. Thus our abstract semidirect factor is exactly the source's actual group of diagram automorphisms.

Conjugation by p interchanges a_1, a_2, whence

$$H_2 = \langle x, p \rangle = \langle a_1, a_2 \rangle \rtimes \langle p \rangle \quad \text{has order 12.}$$

Its involutory generators have product of order six: $(xp)^2 = a_1 a_2$ has order three and the B projection of xp has order two. Also $H_0 = \langle p, q \rangle = B$ has order eight, and $H_1 = \langle x, q \rangle \cong C_2^2$, since q fixes a_1 and x is nontrivial in the base. Semidirect coordinates give

$$H_0 \cap H_2 = \langle p \rangle, \qquad H_0 \cap H_1 = \langle q \rangle, \qquad H_1 \cap H_2 = \langle x \rangle. \tag{4}$$

For the last equality the projections lie in $\langle q \rangle \cap \langle p \rangle = 1$; the base part of H_1 is exactly $\langle x \rangle$. The other two equalities are immediate from the two-factor coordinates. All named generators are distinct, so these identities include the complete intersection property for the three-generator marking, with Borel subgroup one. By (3) these are precisely the source's maximal parabolics, reordered as (H_0, H_1, H_2).

We verify flag transitivity directly. Take three pairwise incident cosets, one of each type. Translate a common representative of the H_0 and H_2 cosets to the identity. The remaining coset gH_1 meets H_0 at some u and H_2 at some v. Then $v^{-1}u \in H_1$, so $v^{-1}u = x^{\epsilon} q^{\delta}$. The element $vx^{\epsilon} = uq^{-\delta}$ lies in all three cosets. Thus every flag has a common representative, is a translate of a standard flag, and extends to a chamber. Each rank-two residue has acting group one of H_0, H_1, H_2, and its two child groups in (4) generate it. Its incidence graph is connected: a product of parabolic elements gives a path from the standard flag. The whole incidence graph is connected because the maximal parabolics generate G_s. This proves residual connectedness. Finally every minimal parabolic has order two and the Borel is one, so all panels have size two. The regular chamber action is faithful; an automorphism fixing a chamber fixes each adjacent chamber in a thin geometry and then all chambers by connectedness. Thus it is also the full type-preserving automorphism group. □

5.4 The input failure and the finite model

The four maximal proper parabolics of W fix, respectively,

$$c_1 = (3, -1, -1, -1)/4, \quad c_2 = (1, 1, -1, -1)/2, \quad c_3 = 0, \quad c_4 = (1, 1, 1, -3)/4.$$

Their linear parts are S_4, and an element fixing c_i has translation $c_i - \pi c_i$. At $s = 1$, the intersection of the c_2 and 0 stabilizers modulo M is the order-eight group of linear permutations preserving or interchanging the blocks $\{1, 2\}, \{3, 4\}$. This follows because $c_2 - \pi c_2 \in M$ exactly in those two cases. Intersecting additionally with the c_1 stabilizer forces coordinate 1 to be fixed, leaving $\langle (34) \rangle$; intersecting with the c_4 stabilizer forces coordinate 4 to be fixed, leaving $\langle (12) \rangle$. Nonzero differences $c_1 - \pi c_1$

and $c_4 - \pi c_4$ have exactly two odd coordinates and hence cannot lie in M. These two child groups generate only a group of order four, so the rank-two residue of the flag of types 2 and 3 is disconnected. The direct proof above shows that this input defect disappears in the particular twisting construction.

The supplementary integer program constructs all four quotient maximal parabolics and their actual intersections, and reconstructs the order-96 quotient, the eight literal diagram automorphisms and their composition, and the order-768 semidirect product. It compares all three literal parabolics, all 64 generated-subgroup intersections and 343 proper product identities. The full helper is retained alongside it; none of its other finite parameter checks is a premise of the theorem. There is no Lean claim. The exact source assertion being corrected is the page-46 conclusion that the $s = 1$ twist is not a regular hypertope. This is a version-bound theorem/example correction, not a numbered-conjecture solution, a full-source admission or a claim of historical priority.

6 Graphs of geometries and the boundary of iteration

A particular fundamental-group presentation can be compared by explicit changes of basis. This does not prove that arbitrary geometric HNN steps remain admissible in succession. The source's three-tree example has the exact positive comparison below; the subsequent two-loop example isolates the obstruction to an unrestricted iterative recipe.

6.1 The three spanning trees in Example 5.18

Use the six source generators and stable letters t_G, t_H, t_K. For the first tree the relations are

$$a_1 = b_4, \quad b_3 = c_6, \quad t_G^{-1} c_5 t_G = a_2, \quad t_G^{-1} c_6 t_G = a_1.$$

Eliminate four names, obtaining the free basis $x = b_3 = c_6$, $y = c_5$, $t = t_G$, with $a_1 = b_4 = t^{-1}xt$ and $a_2 = t^{-1}yt$. For the second tree use $x = a_1 = b_4 = c_6$, $y = a_2 = c_5$, $t = t_H$, and eliminate $b_3 = t^{-1}xt$. For the third use $x = a_1 = c_6 = b_3$, $y = a_2 = c_5$, $t = t_K$, and eliminate $b_4 = t^{-1}xt$. Every substitution is reversible and leaves no relator on x, y, t; these are explicit isomorphisms onto $F(x, y, t)$.

For each tree the final types are $X = \{1, 3, 4, 6\}$, $Y = \{2, 5\}$ and the stable type. Their maximal parabolics become

$$\langle y, t\rangle, \qquad \langle x, t\rangle, \qquad \langle x, y\rangle.$$

For instance the first tree's X subgroup is $\langle a_2, c_5, t_G\rangle = \langle t^{-1}yt, y, t\rangle = \langle y, t\rangle$, and its stable subgroup is $\langle b_3, c_5, c_6\rangle = \langle x, y\rangle$. All others follow from the displayed substitutions. Thus the same group isomorphisms identify the actual parabolics and induce coset incidence isomorphisms, establishing the full example. Each is the standard rank-three free-group geometry, with infinite panels. On source page 56, $a_2 = c_5$ is incorrectly claimed also for the first tree; there they are conjugate. Merely finding separately isomorphic parabolic groups would not suffice for an incidence isomorphism, whereas the explicit marked maps do.

This example does not settle the general fundamental-geometry or universal-property questions in Sections 5.4–6. The obstruction below concerns only the naive sequential algorithm. All source application attributions remain with Piedade and Tranchida; no historical-priority or new-question claim is made here.

6.2 Failure of sequential admissibility

Piedade–Tranchida, arXiv:2505.24662v1, Section 5.4, proposes graphs of coset incidence systems and asks whether the amalgamation and HNN operations give a fundamental geometry independent of the spanning tree. The following example isolates an earlier obstruction: locally admissible edge maps need not remain admissible after the first HNN operation. It concerns the stated iterative procedure, not the impossibility of defining a different fundamental geometry.

Proposition 9. *There is a graph with one vertex and two loops, all of whose vertex and edge geometries are the same finite regular hypertope and all of whose edge identifications are admissible, for which neither ordering of the two prescribed geometric HNN operations is admissible throughout.*

Proof. Let $A = C_2^3$ with basis e_1, e_2, e_3, and give it the standard coset geometry with maximal parabolics

$$A_i = \langle e_j \,:\, j \neq i \rangle \quad (i = 1, 2, 3).$$

This is the join of three two-point geometries, hence is thin, residually connected and flag-transitive, with trivial Borel. The graph has this geometry at its sole vertex and at both loops. On each loop take one boundary map to be the identity. Take the other to be respectively the coordinate transpositions $\sigma = (12)$ and $\tau = (23)$. These are actual automorphisms permuting every indexed parabolic, so each edge map is admissible.

Perform the σ operation first and call its stable letter t. The resulting group is $G = A \rtimes_\sigma \langle t \rangle$, with t of infinite order. The type classes and maximal parabolics prescribed by the source construction are

$$\{1, 2\}, \quad \{3\}, \quad *, \qquad G_{\{1,2\}} = \langle e_3, t \rangle, \quad G_{\{3\}} = \langle e_1, e_2, t \rangle, \quad G_* = A.$$

The standard parabolics contained in A are exactly

$$A, \qquad \langle e_3 \rangle, \qquad \langle e_1, e_2 \rangle, \qquad 1.$$

Indeed every intersection not involving G_* contains the infinite-order element t, and the intersections involving it are precisely the four groups displayed. The second edge still identifies the parabolic A with itself by τ, but

$$\tau(\langle e_3 \rangle) = \langle e_2 \rangle$$

is not on this list. Thus τ is not admissible for the new geometry.

If the τ operation is performed first, its two old type classes are $\{1\}$ and $\{2, 3\}$. The proper nontrivial parabolics in its A residue are $\langle e_1 \rangle$ and $\langle e_2, e_3 \rangle$. The remaining map σ sends $\langle e_1 \rangle$ to $\langle e_2 \rangle$, again absent. Hence the reverse ordering also fails. Both transpositions are their own inverses, so reversing loop orientation does not repair either order. □

The abstract fundamental group still exists: it has presentation

$$\langle A, t, u \mid t^{-1} a t = \sigma(a),\ u^{-1} a u = \tau(a)\ (a \in A) \rangle \cong A \rtimes F(t, u).$$

The obstruction is in the parabolic geometry retained after one operation. A simultaneous construction, a common refinement of the retained incidence data, or a global choice of type classes might avoid it; none is supplied by this example. The source question must therefore distinguish an unrestricted iterative recipe from a construction with additional coherent type data. This limits the proposed iterative recipe; it is not an answer to the broader existence question.

7 Proof boundary and reproducibility

All arguments used above are present either here or in the complete common paper. The additional retained inputs are the faithful affine Coxeter realization and the classical geometric reflection interpretation of the triangle examples. For the affine Weyl semidirect description an exact reference is P. Etingof and X. Ma, *Lecture notes on Cherednik algebras*, Section 7.12, page 58; faithfulness of the standard Coxeter presentation remains the classical affine reflection theorem. The local lattice and quotient computations are proved above.

The complete integer program `affine_example.py` is a proof premise for precisely $s = 2, 3, 4$, checking all 675 parabolic intersections and 10,125 product identities in those cases. The uniform argument for $s \geq 5$ is written. The complete programs `presentation_identities.py` and

`literal_twist.py` are validation: the presentation inverse maps and the all-s twist theorem have independent written proofs. Their full files, with this distinct role, form the program appendix; no search trace is used to replace a proof. The operational checkers and existing partial Lean evidence are inventoried separately. No new formalization is claimed.

The general existence of a fundamental geometry independent of all choices, and categorical universal-property or naturality statements, are not proved here. The two-loop obstruction concerns a specified sequential recipe. Several nontrivial type orbits are already covered by the later general common orbit theorem and receive no new question credit. Unneeded RC1-only auxiliary claims outside the stated flag-transitive hypotheses are not silently certified. The integrated edition awaits a review of its migration and presentation, separate from the accepted complete precursor proofs.

References

[1] C. A. Piedade and P. Tranchida, *From Group Operations to Geometric Structures: Amalgamations, HNN-Extensions, and Twisting in Coset Geometries*, arXiv:2505.24662v1 (2025).

[2] Human Compression Project, *Coset intersections under group operations*, complete common paper accompanying this application.